

Differentiating Accumulation Functions with the FTC

Answer Key

AP Calculus AB/BC — Integration

Quick Answers

1. 29

2. 95

3. $4\sqrt{65}$

4. —

5. $-\log(5)$

6. 70, 2

7. $t^4 - 3t^2$, $4x^3 - 6x$

8. $\frac{6}{x}$, 3

1. $g(x) = \int_1^x (t^3 + 2) dt$ has a constant lower limit and the variable alone on top, so the FTC applies directly: $g'(x) = x^3 + 2$ — the integrand with t replaced by x . Evaluating the integral first is legal but wasted work.

$$g'(3) = 3^3 + 2 = 27 + 2$$

$g'(3) = 29$

2. Here the question asks for accumulated volume, not a rate, so the integral must actually be evaluated (FTC Part 2). An antiderivative of $6t + 4$ is $3t^2 + 4t$:

$$V(5) = \int_0^5 (6t + 4) dt = \left[3t^2 + 4t \right]_0^5 = (75 + 20) - 0$$

95 L

Units check: (litres per minute) $\times$ (minutes) = litres, which is why an area under a rate graph is a volume.

3. The upper limit is x^2 , not x , so this is the chain-rule form: if $g(x) = \int_a^{u(x)} f(t) dt$ then $g'(x) = f(u(x)) \cdot u'(x)$.

$$g'(x) = \sqrt{1 + (x^2)^3} \cdot 2x = 2x\sqrt{1 + x^6}$$

$$g'(2) = 2(2)\sqrt{1 + 2^6} = 4\sqrt{65} \approx 32.25$$

$g'(2) = 4\sqrt{65}$

Forgetting the factor $u'(x) = 2x$ gives $\sqrt{65} \approx 8.06$, a quarter of the correct value — the single most common slip on this problem type.

4. Open response — graded on the justification. A complete answer uses $g'(x) = f(x)$ (FTC) three times:

Increasing. On $[0, 3]$ the table shows $f(t) \geq 0$ (values 4, 3, 2, 0), so $g'(x) = f(x) \geq 0$ there and g is increasing on $[0, 3]$.

Decreasing. On $[3, 7]$ the table shows $f(t) \leq 0$ (values 0, -2, -3, -4, -5), so $g'(x) = f(x) \leq 0$ there and g is decreasing on $[3, 7]$.

Maximum. g' changes sign from positive to negative at $x = 3$, and g is continuous, so by the first-derivative test g has its absolute maximum on $[0, 7]$ at $x = 3$. (Because g increases up to $x = 3$ and decreases afterwards, no endpoint can beat it.)

Accept an answer that says “ g is increasing where f is above the axis” provided it names $g' = f$ as the reason. Do *not* accept “ g has a maximum at $x = 3$ because f has a zero there” with no sign analysis: a zero of f with no sign change would give no extremum. Note also that f decreasing means g is concave down throughout — a good follow-up question.

5. The variable sits in the **lower** limit, so flip the limits first — that introduces a minus sign, and the FTC then applies to the flipped form:

$$h(x) = \int_x^4 \ln(t^2 + 1) dt = - \int_4^x \ln(t^2 + 1) dt \implies h'(x) = -\ln(x^2 + 1)$$

$$h'(2) = -\ln(2^2 + 1) = -\ln 5 \approx -1.609$$

$$h'(2) = -\ln 5$$

The sign is the whole point: as x increases, the interval $[x, 4]$ shrinks, so an accumulation with a variable lower limit *decreases* whenever the integrand is positive.

6. (a) Accumulated volume needs the integral evaluated. An antiderivative of $12 - t$ is $12t - \frac{1}{2}t^2$:

$$V(10) = \left[12t - \frac{1}{2}t^2 \right]_0^{10} = 120 - 50$$

so 70 litres are added in the first 10 minutes.

70

(b) V is an accumulation function of r , so the FTC gives $V'(x) = r(x)$ with no work: $V'(10) = r(10) = 12 - 10$, i.e. 2 litres per minute. At $t = 10$ minutes the reservoir is still gaining water, but only at 2 L/min — the inflow rate has fallen to a sixth of its starting value of 12 L/min.

2

7. (a) Evaluate with FTC Part 2. An antiderivative of $4t^3 - 6t$ is $t^4 - 3t^2$, and it vanishes at $t = 0$:

$$g(x) = \left[t^4 - 3t^2 \right]_0^x = x^4 - 3x^2 - 0$$

$$g(x) = x^4 - 3x^2$$

(b) Differentiating that polynomial gives $4x^3 - 6x$, which is exactly the integrand with t replaced by x — precisely what the FTC predicted, since $g(x) = \int_0^x f(t) dt$ forces $g' = f$.

$$g'(x) = 4x^3 - 6x$$

This problem is the theorem in miniature: integrating and then differentiating returns the original function.

8. (a) Chain-rule form with $u(x) = x^2$ and $f(t) = \frac{3}{t}$:

$$g'(x) = f(x^2) \cdot \frac{d}{dx}(x^2) = \frac{3}{x^2} \cdot 2x = \frac{6x}{x^2}$$

which simplifies to

$$g'(x) = \frac{6}{x}$$

(b) $g'(2) = \frac{6}{2}$.

$$g'(2) = 3$$

Worth showing the class: evaluating first gives $g(x) = 3 \ln(x^2) = 6 \ln x$, whose derivative is again $6/x$. Two routes, one answer — the chain-rule form is simply the one that does not require the integral to be doable.